OMIS 34 – Science, Information Technology, Business and Society

Winter 2026, Jan 5 – Mar 20

Class Time and Location: Section 4 T Th 3:50 – 5:30pm Vari Hall 129

Section 5 T Th 8:30 – 10:10am Lucas Hall 310

Section 6 T Th 12:10 – 1:50pm Vari Hall 135

Professor Yaqiong (Joan) Wang

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Office Hour: Tuesday 2:30 - 3:30pm or by appointment

Office Location: Lucas Hall 316L or zoom

COURSE OVERVIEW

This course examines the complex relationship between science, information technology, business, and society. Through a combination of lectures, readings, case studies, exercises, and projects, we will explore the following questions:

- What are the major breakthroughs in information technology?
- How were these breakthroughs adopted by business and how did they affect business and society?
- What are the social and cultural values in business, science and technology?
- What are the economic challenges posted by rapid business-related information technology?
- What are the workings of information technologies used in business today?

At the end of this course, students should have a deeper understanding of the complex relationship between science, information technology, business and society and should be able to:

1. Describe major IT development that has profound social and economic impacts. In particular, the scientific breakthroughs that bring about the development, the obstacles that need to be overcome, the socioeconomic constraints that affect the design choices and the impact of these development on business, society as well as information technology progress.
2. Understand the workings of business-related information technologies, e.g., hardware, software, database management systems, enterprise application systems, communication networks, the Internet, etc., and how they can be used to solve various business problems.
3. Understand a set of techniques (e.g., customer relationship management, enterprise resource planning, supply chain management, etc.) and principles (e.g., regarding market data usage, financial considerations, business competition and company infrastructure) applied in enterprises to facilitate efficient and effective business operations.
4. Describe individual jobs and IT tools used for operations, finance, accounting, marketing, sales, IT, logistics, human resources, internal communications, database management, etc. within a company.

INSTRUCTOR INFORMATION

I am an Assistant Professor at Leavey School of Business, studying recommender systems, application and privacy of AI. I earned my PhD in Information Systems from University of Minnesota in 2020. My teaching experiences include MIS major classes, e.g., predictive analytics (during my PhD), business driven technology and introduction to programming with python.

I love teaching and believe in active learning. In my class, in addition to transmitting knowledge, I also try various ways to create a free and stimulating environment for my students to take charge and independently construct their own knowledge. If you have any questions or need any help, please feel free to reach out to me. Email is the best way to reach me, and I typically respond within 24 hours.

COURSE MATERIALS

Recommended Textbook: Paige Baltzan. 2026. *Business Driven Technology*, 11th Edition,

McGraw-Hill Education, ISBN 978-1-264-53598-9

Different purchase options.

Additional Materials: Lecture slides, case materials, updates, etc. will be posted on our Camino site:

https://camino.instructure.com/courses/114930

Note that several lectures introduce contents not covered in the textbook. Please check our Camino site regularly for updates.

You are responsible to ensure that your email address listed on Camino is correct and that you do not miss important announcements or other information. "Not knowing" is not an acceptable excuse.

COURSE REQUIREMENTS

1. Class Attendance/Participation (10 points)

Class attendance and participation are required. This portion of your grade will consist of:

- Attending lectures
- The instructor knowing your face and name
- Answering questions raised by the instructor
- Asking interesting questions in class

Throughout the quarter, there will be attendance and attention checks through iClicker every week. 0.5 point will be deducted for each absence. Those checks can be on either Tuesday or Thursday.

Attendance exceptions will be granted in rare circumstances only with prior approval. Please let me know as much in advance as possible if there is a need to miss a class.

2. Case Studies (15 points)

In addition to lectures, this class will also include 5 case studies in the format of group discussion. Everyone in a preassigned group is expected to read all case materials prior to class and to actively participate in discussion. This portion of your grade will be awarded based on you and your group's in-class contribution to case discussion (consider both quantity and quality) and 1 or 2-page after-class written summary. Individual grade on case studies will be weighted by peer-reviewed performance on in-class discussion.

Tips for In-Class Case Discussion

In-class case discussion and contribution can have various forms. Groups and students who summarize, clarify,

ask any relevant questions, or suggest what else needs to be covered will all be considered as making a valuable contribution. If you are shy about contributing, consider preparing well and bringing your thoughts or questions on paper. If you are confident about public speaking, focus on increasing the quality-to-airtime ratio of your participation. I encourage you to take some risks in this class, assuming that you are well prepared. I will take many factors into account to evaluate your contribution to class learning.

Examples of things viewed positively:

- good analysis supported by case facts or your own experience
- good arguments expressed coherently and succinctly
- asking interesting questions about assigned readings
- relevant comments that incorporate concepts covered in class
- listening and building on what others say
- summarizing or reconciling previous thread of discussion
- constructive disagreement with rationale
- summarizing main takeaways from a specific case study
- testing and exploring new ideas that enhance learning

I will make every attempt to give groups equal opportunities to participate. However, if you are concerned about your case participation, please don't hesitate to talk to me. I will be happy to discuss your case participation, or any other issue you might have on an individual basis. Obviously, you should participate in class if you want to share your ideas with others. Yet there is no need to try to dominate the discussion. Some of the best contributors participate less often than the most active speakers. Quality is as important as quantity.

3. Practice Quizzes (10 points)

There will be 5 practice quizzes throughout this course. The goal of the quizzes is to help you review the concepts covered in lectures. These quizzes are **open book** and **take-home exercises**. They will typically be posted at the end of certain chapters or units. You will get notices when they are available.

4. Examinations (40 points)

There will be 3 examinations throughout this course, i.e., 2 Exams during the quarter and 1 Final Exam. All these exams are **closed book** and will be taken **in-class**. Each will be comprehensive and cover all materials assigned up to the date of the exam.

Makeup tests will not be given unless for serious legitimate excuses allowed by university policies and arranged in advance.

5. Team Project (25 points)

Students will also be required to work on a team project throughout the course. The goal of this assignment is to conduct an in-depth investigation of a specific challenge or topic related to IT and its business applications within the context of a company. There is a list of companies in various industries to choose from. Students will prepare a **project report** and deliver a **15-minute presentation** in class. The report and presentation will be evaluated on three criteria: depth of research and analysis, presentation structure and clarity, and public speaking skills including engaging the audience.

More details (timeline, deliverables, grading) about the team project are elaborated in Team Project Guide.

GRADING POLICY

Your final grade in this course will be based on the following.

	Points	Comments
Class Attendance and Participation	**10**	
Attendance and general participation	10	0.5 point for each attendance check; In class Q&A
Case Studies	**15**	
Case discussion	5	3 points for each case study
Practice Quizzes	**10**	
Take-home exercises	5	2 points for each practice quiz
Examinations	**40**	
Exam#1	10	Chapters: B1, 1, 2, 3
Exam#2	10	Chapters: E1, 6, 7, 8, 14, 16
Final Exam	20	**Comprehensive – ALL materials covered**
Team Project	**25**	
Project Outline	5	Team deliverable
Team Presentation	10	Slides and contents delivery
Written Report	10	Team deliverable and individual participation
Total	**100**	

Note that I reserve the right to change the distribution and add assignments as necessary, as a result, the final total grade may vary a bit as events of the quarter unfold and affect some assignments.

Your final letter grade in this course will be based on **curved credits** and follow the distribution below:

A	**93-100%**	**B**	**83-86%**	**C**	**73-76%**	**F**	**0-59%**
A-	**90-92%**	**B-**	**80-82%**	**C-**	**70-72%**		
B+	**87-89%**	**C+**	**77-79%**	**D**	**60-69%**		

Extra Credit(s) Opportunities

3 extra credits in total will be awarded to encourage lecture participation, course feedback provision and student presentation engagement. These extra credits will be awarded as follows:

- Students who ask or answer questions during the lectures in each session will be awarded **0.2 extra credit**. Up to **1.5 extra credits** will be awarded throughout the course.
- Every time a student engages with team project presentation, for example asking questions or giving comments, will be awarded **0.3 extra credit**. Up to **1 extra credit** will be awarded throughout the project presentations.
- Students who complete the final course evaluation will be awarded **0.5 extra credit**.

Policies on Late Assignments

All writing assignments will have specific due dates. Students are required to submit an electronic copy in word or pdf formats on Camino.

Late assignments will incur a 10% penalty per day unless it is due to a problem that is beyond your control, e.g., medical or business related. Specifically excluded are personal and vacation plans. Contact the instructor as soon as you know that you have a conflict.

CLASSROOM COMPUTER USE

Cell phones, pagers, and similar electronic communication devices should be turned off during all classes. If you have an urgent phone call or text to respond to, please quietly exit the classroom and take care of it outside of our classroom or offline. You are welcome to use notebook computers or other devices to take notes as long as it does not hinder your ability to actively participate in class discussion.

ACADEMIC POLICY

Student responsibility: By enrolling in this class, a student is bound by the requirements stated in this syllabus and given by the instructor. Students must operate with integrity in their dealings with faculty and other students; engage the learning materials with appropriate attention and dedication; maintain their engagement when challenged by difficult learning activities; contribute to the learning of others; and perform to standards set by the faculty.

Classroom behavior: Students should behave properly during class time to facilitate the teaching and learning.

ACADEMIC INTEGRITY

The Academic Integrity pledge is an expression of the University's commitment to fostering an understanding of -- and commitment to -- a culture of integrity at Santa Clara University. The Academic Integrity pledge, which applies to all students, states:

I am committed to be a person of integrity. I pledge, as a member of the Santa Clara University community, to abide by and uphold the standards of academic integrity contained in the Student Conduct Code.

Students are expected to uphold the principles of this pledge for all work in this class. For more information about Santa Clara University's academic integrity pledge and resources about ensuring academic integrity in your work, see www.scu.edu/academic-integrity.

Work done for this course must adhere to the University's expectations outlined in the University Bulletin and the Student Handbook. The minimum penalty for cheating on an assignment is a failing grade. Cheating on exams will result in an F for the course. In addition, cheating will also be reported to the Office of Student Life according to the University Academic Integrity Protocol.

OFFICE OF ACCESSIBLE EDUCATION (FORMERLY DISABILITY RESOURCES)

If you have a documented disability for which accommodations may be required in this class, please contact the Office of Accessible Education (Benson 1, http://www.scu.edu/oae*, 408-554-4109) as soon as possible to discuss your needs and register for accommodations with the University. If you have already arranged accommodations through OAE, please discuss them with me during my office hours during the first two weeks of class.*

To ensure fairness and consistency, individual faculty members are required to receive verification from the Office of Accessible Education before providing accommodations. OAE will work with students and faculty to arrange proctored exams for students whose accommodations include double time for exams and/or assisted technology. Students with approved accommodations of time-and-a-half should talk with me as soon as possible.

OAE must be contacted in advance (at least two weeks of notice recommended) to schedule proctored examinations or to arrange other accommodations.

ACCOMMODATIONS FOR PREGNANT AND PARENTING STUDENTS

In alignment with Title IX of the Education Amendments of 1972, and with the California Education Code, Section 66281.7, Santa Clara University provides reasonable accommodations to students who are pregnant, have recently experienced childbirth, and/or have medically related needs related to childbirth. Pregnant and parenting students can often arrange accommodations by working directly with their instructors, supervisors, or departments. In addition, the Office of Accessible Education will provide reasonable accommodations for pregnancy-related impairments which impact a major life activity.

DISCRIMINATION AND SEXUAL MISCONDUCT (TITLE IX)

Santa Clara University upholds a zero-tolerance policy for discrimination, harassment and sexual misconduct. If you (or someone you know) have experienced discrimination or harassment, including sexual assault, domestic/dating violence, or stalking, I encourage you to tell someone promptly. For more information, please consult the University's Gender-Based Discrimination and Sexual Misconduct Policy at http://bit.ly/2ce1hBb *or contact the University's EEO and Title IX Coordinator, Belinda Guthrie, at 408-554-3043,* bguthrie@scu.edu. *Reports may be submitted online through* https://www.scu.edu/osl/report/ *or anonymously through Ethicspoint* https://www.scu.edu/hr/quick-links/ethicspoint/

IN-CLASS RECORDINGS

The Student Conduct Code (p. 13) prohibits students from "making a video recording, audio recording, or streaming audio/video of private, non-public conversations and/or meetings, inclusive of the classroom setting, without the knowledge and consent of all recorded parties," except in cases of approved disability accommodations. The Student Conduct Code also prohibits the "falsification or misuse, including non-authentic, altered, or fraudulent misuse, of University records, permits, documents, communication equipment, or identification cards and government-issued documents." Dissemination or sharing of any classroom recording without the permission of the instructor would be considered "misuse" and, therefore, prohibited. Violations of these policies may result in disciplinary action by the University. At the instructor's discretion, violations may also have an adverse effect on the student's grade.

ASSOCIATION WITH THE "VALUES IN SCIENCE AND TECHNOLOGY" PATHWAY

You can find information about Pathways on the Core Curriculum website https://www.scu.edu/provost/core/ *including specific Pathways, all courses associated with them, and the Reflection Essay prompt and rubric used to evaluate the final essay you will submit.* https://www.scu.edu/provost/core/pathways/pathway-reflection-essay-instructions/essay-submissions/

If you declare this Pathway, you may use a representative piece of work from this course as one of the Pathways materials you will upload via e-Campus during your junior or senior year. Therefore, we recommend that you keep electronic copies of your work using Dropbox or Google Docs, in addition to saving copies on your own computer or flash drives. This may ensure you will have a range of choices for retrieving your saved files when you analyze and assemble your Pathways materials in preparation to write the Pathways reflection essay.

TENTITIVE CLASS SCHEDULE

This schedule is subject to change with fair notice. Notification of any changes will be sent to all students.

Session #	Class Date - Day	Topics / Readings	Assignments and Class Activities
1	Jan 6 - Tu	➢ Course Overview ➢ Syllabus Walkthrough ➢ Team Project Guide	*Students check the recommended Textbook;* *Students read recommended materials before class.*
2	Jan 8 - Th	➢ **Chapter B1:** Business Basics	
3	Jan 13 - Tu	➢ **Chapter 1:** Business Driven Technology	
4	Jan 15 - Th	➢ **Chapter 2:** Identifying Competitive Advantages ➢ **Case Discussion**	***Case Study#1*: Big Data, Big Analytics** *Readings and questions posted on Camino.*
5	Jan 20 - Tu	➢ **Chapter 2**…continued ➢ **Chapter 3:** Implementing Competitive Advantages	***Team Project*:** ➢ *One team member emails **top 3 company choices** to me (cc all members)* ➢ *I will approve companies* *Each team will study a different company*
6	Jan 22 - Th	➢ **Chapter 3**…continued ➢ **Chapter E1:** Computer / Internet History	***Practice Quiz#1*: ChB1, 1, 2, 3**
7	Jan 27 - Tu	➢ **Chapter E1**…continued ➢ **Exam#1**	**EXAM#1: Chapters B1, 1, 2, 3**
8	Jan 29 - Th	➢ **Chapter 16:** Wireless Technology ➢ **Case Discussion**	***Case Study#2*: Internet History** *Readings/video link and questions posted on Camino.*
9	Feb 3 - Tu	➢ **Chapter 14:** E-business ➢ **Exam#1 Review**	***Practice Quiz#2*: ChE1, 14, 16**
10	Feb 5 - Th	➢ **Chapter 6**: Valuing and Storing Organizational Information – Databases ➢ **Case Discussion**	***Case Study#3*: Blockchain** *Readings and questions /video link posted on Camino.* ***Team Project (preliminary findings): two-page outline including*** *1) Company description* *2) Business processes of interest* *2) IT/IS in selected business processes* *3) Problem(s) and/or opportunity(ies)*
11	Feb 10 - Tu	➢ **Chapter 6**…continued ➢ **Chapter 7:** Accessing Organizational Information – Data Warehouses	
12	Feb 12 - Th	➢ **Chapter 8**: Understanding Big Data and Its Impact on Business	
13	Feb 17 - Tu	➢ **Chapter 8**…continued ➢ **Chapter E2:** AI and Neural Networks	***Practice Quiz#3*: Ch6, 7, 8**

Session #	Class Date - Day	Topics / Readings	Assignments and Class Activities
14	Feb 19 - Th	➢ **Chapter E2**…continued ➢ **Case discussion**	***Case Study#4*: Generative AI** *Readings/video link and questions posted on Camino.*
15	Feb 24 - Tu	➢ **Chapter 10**: Supply Chain Management (SCM) ➢ **Exam#2**	**EXAM#2: Chapters E1, 16, 14, 6, 7, 8**
16	Feb 26 - Th	➢ **Chapter 10**…continued ➢ **Chapter 11:** Customer Relationship Management (CRM) ➢ **Exam#2 Review**	***Team Project* (Discussion with Professor):** *No later than this date each Team should have met me ONCE to discuss project progress.* ***Practice Quiz#4*: ChE2, 10**
17	Mar 3 - Tu	➢ **Chapter 11** …continued ➢ **Chapter 12**: Enterprise Resource Planning (ERP)	
18	Mar 5 - Th	➢ **Chapter 12**…continued ➢ **Case discussion**	***Case Study#5*: Cloud Computing** *Readings and questions posted on Camino.* ***Practice Quiz#5*: Ch11, 12**
19	Mar 10 - Tu	➢ **Team Presentations** ➢ Final Exam Study Guide ➢ Exam Date/Location Announcement	Being absent for other team presentations will **deduct 5 points from your project grade** **Extra Credit** - Questions during presentations
20	Mar 12 - Th	➢ **Team Presentations** ➢ **Course Review**	
	Mar 19 - Th	➢ **FINAL EXAM** Sec350: 6:30pm, Mar 19, 129 Vari Hall Sec830: 9:10am, Mar 19, 310 Lucas Hall Sec1210: 1:30pm, Mar 19, 135 Vari Hall	***Team Project* (Final Report due by 5p Friday):** ***Project report due in Camino (one per Team)*** ***FINAL EXAM - Chapters 1, 2, 3, 6, 7, 8, 10, 11, 12, 14, 16, B1, E1, E2 handout, cases, class material & discussions.***